Case Study Summary: Exploring Customer Behavior Through Database Analysis

1. Executive Summary & Introduction

This case study analyzes an online food ordering platform's data ecosystem to uncover deep insights into customer behavior, visiting patterns, spending habits, and premium loyalty program performance. By exploring relations between multiple tables in the database, the analysis provides actionable intelligence to assist with data-driven decisions regarding marketing strategies, platform enhancements, and expansion plans for the customer loyalty program.

2. Database Architecture & Key Datasets

The analysis integrates five fundamental relational tables:
• **Sales:** Details regarding specific transactions, order dates, user IDs, and product purchases (62 records).
• **Product:** Menu offerings, names (e.g., Butter Chicken, Dal Makhani), and corresponding standard prices (11 offerings).
• **User Name:** References matching User IDs with distinct, human-readable fictional customer names.
• **Users:** System-wide initial account creation dates ranging from 2013 to 2016.
• **Goldusers_Signup:** Premium membership registration tracking records detailing when users upgraded to the Gold tier.

3. Summary of Insights & Business Impact

• **Revenue Optimization:** High-revenue dishes such as Butter Chicken can be targeted heavily in promotions to maximize gains, while underperforming menu choices can be optimized or removed.
• **Loyalty Metrics:** Quantifying spending patterns before and after premium membership helps evaluate customer lifetime value (LTV) and determine the exact return on investment (ROI) of the Gold tier loyalty initiative.
• **Operational Efficiency:** Providing pre-aggregated analytical datasets empowers internal business squads to run periodic performance inspections without requiring continuous manual SQL coding support.